

Keeping It Simple: Birth Seasonality in the Age of AI

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Monthly birth counts exhibit recurring seasonal variation associated with biological, behavioural, climatic, cultural, and institutional processes. Forecasts of these fluctuations inform maternity and neonatal service planning and demographic projection. Forecasting methods increasingly incorporate neural architectures, offering greater representational capacity alongside increased model complexity and computational overhead. These trade-offs raise the question of whether additional capacity improves predictive accuracy for demographic series with identifiable trend and seasonal components. We compare 14 statistical, neural, and hybrid models using 228 monthly observations of births in the United Kingdom from 2006 through 2024. Models are trained through December 2019 and evaluated over a 60-month fixed-origin holdout period covering 2020–2024. The leading statistical model achieves the lowest root mean squared error, whereas selected hybrids yield lower mean absolute error. These findings indicate that neural architectures provide no consistent predictive advantage over explicit statistical modelling of trend and seasonality in this series during the observed evaluation period.